

Structural Prominence in a Joint EmotionCognition Network of Affective Disorders in Adolescents and Young Adults: Stress, Sleep, and Cognitive Performance

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Abstract

Affective symptoms, perceived stress, sleep disturbance, and cognitive dysfunction frequently co-occur in adolescents and young adults with affective disorders, yet the structural organisation of these domains within a joint association network remains unclear. We estimated a cross-sectional partial-correlation network in 1,027 participants and applied domination-based structural-positioning metrics, including minimum dominating sets, control frequency, and average module control strength, to quantify cross-domain structural coverage. We additionally examined cross-sectional associations between RBANS Total Scale score and symptom burden using covariate-adjusted linear regression and tested baseline-adjusted longitudinal associations in a paired cohort of 1,056 participants

with at least two assessments. Negative stress and sleep-related indicators showed relatively prominent cross-domain structural positions within the estimated network. Cross-sectionally, better cognitive performance was associated most consistently with lower depressive and anxiety symptom burden. Longitudinally, higher baseline RBANS Total Scale scores predicted lower follow-up perceived stress severity, whereas prospective associations with other outcomes were limited. These findings support an exploratory structural-prominence account in which stress and sleep are prominently embedded within a broader emotion-cognition system, while cognitive performance is linked to both current affective burden and later perceived stress. The results should be interpreted as hypothesis-generating rather than as evidence of causal control or intervention efficacy.

Introduction

Affective disorders in adolescents and young adults impose substantial functional burden and often extend beyond core mood symptoms alone. In addition to depression and anxiety, many affected individuals experience clinically meaningful cognitive difficulties, elevated perceived stress, and sleep disturbance, all of which may interfere with academic performance, social development, and everyday functioning [1–9]. These domains rarely occur in isolation. Rather, they tend to co-occur and interact, raising a central question for both clinical science and systems-oriented psychopathology: how are affective symptoms, stress, sleep, and cognition organised within a broader multi-domain symptom system?

Symptom-network approaches provide one useful framework for studying such complexity by representing symptoms and related indicators as nodes and their conditional associations as edges [10–14]. This literature has generated important insights into symptom clustering, comorbidity, and network topology. However, much of this work has focused on descriptive organisation or on position-based indices such as centrality [15–17]. Less attention has been given to a related but distinct question: within a joint multi-domain association network, which nodes or domains occupy structurally prominent po-

sitions that allow relatively broad cross-domain coverage? This question is clinically relevant because structural prominence may help identify components that are especially embedded across domains, while still requiring careful interpretation given that cross-sectional partial-correlation networks remain associational rather than causal [18–20].

To address this gap, the present study applies dominance-based structural metrics from network science to characterise how symptoms and cognitive indicators are positioned within a joint emotioncognition association network. Prior work has highlighted the close coupling of emotion and cognition, as well as the importance of stress- and sleep-related processes to affective and cognitive functioning [8, 21–31]. In the current study, Minimum Dominating Sets (MDS), Control Frequency (CF), and Average Module Control Strength (AMCS) were used to quantify minimal structural coverage and cross-domain placement at the node and module levels [32–35]. Here, the term control is retained only as part of the original metric nomenclature; these measures are not interpreted as evidence of causal control, directional influence, or intervention priority. Rather, they complement conventional centrality-based descriptions by asking which indicators recur in minimal coverage configurations and which domains show relatively broader structural reach within the estimated network.

We organise the study around a two-level framework. At the system level, we ask whether stress- and sleep-related indicators occupy structurally prominent cross-domain positions in a joint emotioncognition association network. At the person level, we ask whether cognitive performance, modelled as a continuous clinical dimension rather than as a subgroup label, is associated with current symptom burden and with subsequent outcomes over a short follow-up interval. Accordingly, the present study addresses three questions. First, do stress- and sleep-related indicators show relatively prominent cross-domain structural positioning compared with other domains? Second, when cognitive performance is modelled continuously, is it cross-sectionally associated with symptom burden across depression, anxiety, perceived stress, and insomnia? Third, in baseline-adjusted longitudinal models, is baseline cognitive performance associated with subsequent outcomes across these same domains? These questions are examined within an

exploratory framework intended to characterise structural prominence in an estimated association network and to relate cognitive performance to current and later clinical burden, rather than to infer causal control or intervention efficacy.

Methods

Participants and Data Source

This study included adolescents and young adults aged 11–24 years with affective disorders recruited from the Early Intervention Department of Nanjing Brain Hospital between September 2020 and November 2024. Participants completed online self-report questionnaires, and cognitive assessments were administered by trained psychiatrists. The cross-sectional analytic sample comprised 1,027 baseline participants whose first available assessment corresponded to Session 1 and whose baseline diagnosis was Major Depressive Disorder (MDD) or Bipolar Disorder (BD). Longitudinal data were drawn from a partially overlapping repeated-assessment dataset in the raw clinical database. At the source-cohort level, 1,112 participants had at least two assessments and a non-missing T2 HAMD-17 score; the operational extraction criterion used for the raw repeated-assessment source dataset, not the analytic denominator for all longitudinal models. Among these, 882 also belonged to the cross-sectional cohort. Based on baseline diagnosis codes in the raw database, this broader repeated-assessment source cohort included 566 participants with MDD, 502 with BD, and a small remainder with other diagnosis codes. For the present study, the longitudinal analytic cohort used in the adjusted regression analyses was defined more restrictively to align with the main analytic framework, requiring age 11–24 years, baseline Session 1, and a baseline diagnosis of MDD or BD. This yielded a paired analytic cohort of 1,056 participants, including 561 with MDD and 495 with BD at baseline. Follow-up intervals were right-skewed, with a median of 22 days (interquartile range, 15–35 days; range, 0–1223 days). Because the longitudinal models were estimated using outcome-specific complete-case data, analytic sample size varied across

outcomes. Diagnoses were established by trained psychiatrists using age-appropriate structured clinical interviews in routine clinical practice, including the K-SADS-PL for participants younger than 18 years and a structured clinical interview workflow for participants aged 18 years or older [36, 37]. All participants provided written informed consent before enrolment. In the current analytic pipeline, treatment exposure was not adjusted for in the longitudinal analyses, and the preprocessing scripts did not explicitly exclude neurodevelopmental disorders, substance-use-related conditions, or neurological/somatic comorbidities that may affect cognitive performance.

Measures and Variable Construction

The study employed various self-report questionnaires to measure the presence and severity of depressive, anxiety, insomnia, and stress symptoms among the participants [38–42].

Patient Health Questionnaire-9 (PHQ-9): The PHQ-9 is a 9-item self-report scale assessing the presence and severity of depressive symptoms over the past two weeks. Each item is rated on a 4-point Likert scale (0 = not at all to 3 = nearly every day), with total scores ranging from 0 to 27. Scores of 5–9, 10–14, and ≥ 15 are considered indicative of mild, moderate, and severe depression, respectively [38].

Generalized Anxiety Disorder-7 (GAD-7): The GAD-7 is a 7-item self-report scale used to assess the presence and severity of anxiety symptoms over the past two weeks. Each item is rated on a 4-point Likert scale (0 = not at all to 3 = nearly every day), with total scores ranging from 0 to 21. Scores of 5–9, 10–14, and ≥ 15 are considered indicative of mild, moderate, and severe anxiety, respectively [39].

Perceived Stress Scale-14 (PSS-14): The PSS-14 is a 14-item self-report scale measuring the degree to which situations in one’s life are appraised as stressful over the past month. Each item is rated on a 5-point Likert scale (0 = never to 4 = very often), with total scores ranging from 0 to 56. Scores of 29–41 and ≥ 42 are considered indicative of moderate and high perceived stress, respectively [40].

Insomnia Severity Index (ISI): The ISI is a 7-item self-report scale used to

measure the severity of insomnia symptoms over the past two weeks. Each item is rated on a 5-point Likert scale (0 = no problem to 4 = very severe problem), with total scores ranging from 0 to 28. Scores of 8–14, 15–21, and ≥ 22 are considered indicative of mild, moderate, and severe insomnia, respectively [41].

Repeatable Battery for the Assessment of Neuropsychological Status (RBANS):

The RBANS is a brief, individually administered battery designed to assess cognitive function across five domains: Immediate Memory, Visuospatial/Constructional, Language, Attention, and Delayed Memory. It consists of 12 subtests, yielding index scores for each domain, as well as a Total Scale score. The RBANS was standardized on a normative sample of individuals aged 20–89 years, with a mean Total Scale score of 100 (SD = 15). For the present study, raw scores were used for all subtests [42].

A total of 49 observed indicators were included in the analytic framework, consisting of 37 symptom items and 12 cognitive indicators. The symptom indicators comprised 9 PHQ-9 items, 7 GAD-7 items, 14 PSS items, and 7 ISI items, whereas the cognitive indicators comprised 12 RBANS subtests. Item-level variables were used in the network analyses. For regression-based analyses, total scores were additionally computed for the PHQ-9, GAD-7, PSS, and ISI by summing all items within each scale. Global cognition was indexed by the RBANS Total Scale score. Age, sex, education, and diagnosis were extracted as covariates for adjusted analyses.

Network estimation

All symptom indicators were standardised (Z-scores). We employed Graphical Lasso with EBIC-guided hyperparameter selection. The candidate range comprised 200 log-spaced α values between 10^{-3} and $10^{-0.7}$. EBIC ($\gamma = 0.5$) identified $\alpha = 0.054$ for the full sample. We fitted the model using `sklearn.covariance.GraphicalLasso` (maximum 2,000 iterations). The mathematical formulation of the Graphical LASSO is as follows [43–52]:

$$\hat{K} = \arg \min_K \text{tr}(SK) - \log \det(K) + \alpha \|K\|_1$$

where K represents the estimated sparse inverse covariance matrix, S is the sample covariance matrix, $\|K\|_1$ denotes the sum of the absolute values of the off-diagonal elements of K , and α is the sparsity parameter, set to 0.054 in this study. Implementations were performed using the Scikit-learn library in Python. To focus on the magnitude of associations, partial correlation coefficients within the inverse covariance matrix were absoluteized, and diagonal elements were set to zero, thus isolating inter-node relationships. An edge between nodes i and j was deemed to exist if the corresponding matrix element was non-zero, with the edge weight reflecting the absolute value of that element [43, 46].

Community Detection and Structural Positioning Metrics

We applied the Louvain algorithm to the absolute-weight network (resolution = 1.0, random state = 42) using python-louvain to maximise modularity [53]. Six communities were identified: Anxiety Related Module (ARM), Depression Related Module (DRM), Cognition Related Module (CRM), Negative Stress Module (NSM), Positive Stress Module (PSM), and Sleep Related Module (SRM) [54].

To quantify whether specific indicators occupy structural positions consistent with cross-domain regulation in the joint symptom system, we used dominance-based structural positioning metrics. For this analysis, we treated the estimated symptom network as an undirected, unweighted graph $G = (V, E)$, where nodes represent indicators and an edge indicates a non-zero partial correlation estimated by Graphical Lasso. A set $D \subseteq V$ is a dominating set if every node in V is either in D or adjacent to at least one node in D . Minimum dominating sets (MDSs) are dominating sets with minimum cardinality $k_{\min}$ [32, 33, 55, 56].

We exhaustively enumerated all distinct MDSs using an exact search with pruning. We then computed control frequency (CF) for each node v as the proportion of MDSs

that include v :

$$CF(v) = \frac{1}{|\mathcal{D}|} \sum_{D \in \mathcal{D}} \mathbb{I}(v \in D), \quad (1)$$

where $\mathcal{D}$ denotes the set of all enumerated MDSs [32, 33].

To summarise cross-module structural coverage, for each MDS D we defined the control domain of module i as the union of dominating nodes in module i and their graph neighbours. The AMCS from module i to module j was defined as the average proportion of nodes in module j covered by module i across all MDSs [33]:

$$AMCS_{i \rightarrow j} = \frac{1}{|\mathcal{D}|} \sum_{D \in \mathcal{D}} \frac{|\text{Control}_i^{(D)} \cap V_j|}{|V_j|}. \quad (2)$$

These metrics quantify structural coverage/positioning in the estimated conditional-dependence network and are not interpreted as evidence of causal influence [34, 35].

Continuous Cognitive-Performance Analyses

To examine the cross-sectional association between cognitive performance and symptom burden, we fitted a series of linear regression models with PHQ-9 total score, GAD-7 total score, PSS total score, and ISI total score as separate dependent variables and the RBANS Total Scale score as the predictor of interest. All models adjusted for age, sex, education, and diagnosis. Robust standard errors (HC3) were used for inference, and p values for the RBANS term were corrected for multiple testing across the four outcomes using the false discovery rate (FDR) procedure.

Longitudinal Analyses

The longitudinal regression models were estimated in the analytic cohort defined in the Participants and Data Source section ($n = 1,056$), with model-wise complete-case samples varying by outcome. Follow-up outcomes were modelled using baseline-adjusted linear regression. For each outcome, the latest available assessment was regressed on its baseline

value, baseline RBANS Total Scale score, age, sex, education, diagnosis, and follow-up interval. Heteroskedasticity-consistent HC3 standard errors were used, and false-discovery-rate correction was applied across the five tests of baseline RBANS Total Scale score.

Statistical Considerations

No formal imputation procedure was applied. Missing data were handled using complete-case procedures within the relevant analytic pipeline. For the network analyses, participants with non-missing values on the variables required for network estimation were retained. For regression-based analyses, model-wise complete-case samples were used, such that only participants with complete data on the outcome, predictor, and covariates for a given model were included. All statistical tests were two-sided, and statistical significance was defined as $p < 0.05$ unless otherwise specified. For analyses involving multiple symptom outcomes, p values for the cognitive predictor of interest were adjusted using the false discovery rate (FDR) procedure with the Benjamini-Hochberg method. All analyses were conducted in Python 3.12.10 using NumPy 2.3.5, pandas 2.3.3, SciPy 1.16.3, statsmodels 0.14.6, NetworkX 3.6.1, scikit-learn 1.8.0, matplotlib 3.10.8, and seaborn 0.13.2.

Results

Sample characteristics

Sample characteristics of the cross-sectional and longitudinal analytic cohorts are summarised in Table 1. The longitudinal analytic cohort should be distinguished from the broader repeated-assessment source cohort extracted from the raw database. The cross-sectional analytic sample comprised 1,027 adolescents and young adults with affective disorders (mean age, 15.34 years [SD, 2.03]; 739 females), including 531 participants with MDD and 496 with BD. Cross-sectional continuous regression analyses were conducted

on model-wise complete-case samples of 1,024 participants. The longitudinal adjusted analyses were performed in a paired analytic cohort of 1,056 participants with at least two assessments (mean baseline age, 15.38 years [SD, 2.05]; 760 females), including 561 participants with MDD and 495 with BD at baseline. The median follow-up interval was 22 days (IQR, 15–35 days), and outcome-specific longitudinal sample sizes ranged from 965 to 1,031 participants.

Joint network structure and modular organisation

We first characterised the overall architecture of the joint emotion–cognition association network. Community detection identified six modules aligned with the dominant content domains: Anxiety Related Module (ARM), Depression Related Module (DRM), Cognition Related Module (CRM), Negative Stress Module (NSM), Positive Stress Module (PSM), and Sleep Related Module (SRM) (Figure 1A). A sparse partial-correlation network was estimated across 49 indicators spanning depression (PHQ-9), anxiety (GAD-7), perceived stress (PSS-14), insomnia severity (ISI), and cognitive functioning (RBANS). The full-sample network comprised 49 nodes and 354 edges (density = 0.301), with global efficiency = 0.638 and node-normalised global strength ($\sum_{i < j} |w_{ij}|/N$) = 0.421 (Figure 1B). The network diameter was 3 and the average shortest path length was 1.774, consistent with a compact and highly traversable system. Together, these properties indicate a modular but interconnected emotion–stress–sleep–cognition architecture in which domain-specific clusters remain distinguishable while preserving substantial cross-domain coupling.

To contextualise subsequent structural-positioning analyses, we also summarised multiple node properties, including strength, clustering, closeness, betweenness, degree, and eigenvector centrality, as standardised profiles across nodes (Figure 1C). These profiles indicate marked heterogeneity in how individual indicators are embedded within the broader network topology and provide the descriptive basis for the dominance-based analyses reported below.

Node- and module-level structural positioning

We next asked which indicators and modules occupied the most prominent cross-domain structural positions within the estimated network. Dominance-based analyses revealed a clear multi-domain structural-positioning pattern centred on stress appraisal and sleep disturbance. Exhaustive enumeration of minimum dominating sets (MDSs) identified eight distinct solutions with a minimum size of $k_{min} = 4$, indicating that minimal coverage of the full system consistently involved a small set of indicators spanning multiple domains rather than a single-symptom cluster. Control frequency (CF), defined as the proportion of MDSs in which a node appeared, showed a highly skewed distribution (Figure 1D). The negative stress item PSS12 showed the highest CF (0.750), indicating that it recurred most consistently across minimal coverage configurations. Several additional indicators also showed elevated CF values, including sleep-related, depression-related, and selected cognitive indicators (Figure 2A). Taken together, these node-level results indicate that stress appraisal and sleep symptoms occupy relatively prominent positions in cross-domain structural coverage.

At the module level, three complementary summaries further supported this multi-domain structural-positioning pattern. First, module in-strength and out-strength profiles showed clear asymmetry across domains (Figures 2B–2C): negative stress and sleep ranked among the strongest outbound modules, whereas positive stress showed comparatively greater inbound accumulation. Second, module-aggregated CF identified sleep and negative stress as the most frequently represented modules in minimal coverage solutions (Figure 2D), indicating that repeated cross-domain coverage consistently involved these domains. Third, the directed module-level backbone, quantified using AMCS, highlighted several high-coverage pathways (Figures 2E–2F). Negative stress showed the strongest outbound coverage to positive stress (AMCS = 0.643) and cognition (AMCS = 0.375), whereas sleep showed moderate outbound coverage to anxiety (AMCS = 0.347) and depression (AMCS = 0.312). Anxiety also showed substantial coverage towards depression (AMCS = 0.438), consistent with close affective coupling at the module level. When

modules were further aggregated into higher-level groups, peripheral emotional modules showed stronger outbound coverage towards both intrinsic emotional and cognitive groups (Figure 2F). Overall, these findings support a multi-domain structural-positioning pattern in which stress appraisal and sleep disturbance occupy prominent cross-domain locations linking affective symptoms and cognitive functioning.

Continuous associations between cognitive performance and symptom burden

We then examined whether cognitive performance, modelled as a continuous clinical dimension, was associated with current symptom burden across the full clinical sample. Cross-sectional linear regression models were fitted with RBANS Total Scale score as the predictor of symptom burden while adjusting for age, sex, education, and diagnosis (Figure 3A; Table 2). Higher RBANS Total Scale scores were associated with lower depressive and anxiety symptom burden. Specifically, RBANS total was negatively associated with PHQ-9 total ($\beta = -0.029$, 95% CI $[-0.048, -0.011]$, FDR-adjusted $p = 0.004$) and GAD-7 total ($\beta = -0.037$, 95% CI $[-0.053, -0.021]$, FDR-adjusted $p < 0.001$). Associations with perceived stress and insomnia were weaker. The association with PSS total did not remain significant after multiple-comparison correction ($\beta = -0.026$, 95% CI $[-0.052, 0.000]$, FDR-adjusted $p = 0.053$), and the association with ISI total was likewise attenuated after correction ($\beta = -0.018$, 95% CI $[-0.036, -0.001]$, FDR-adjusted $p = 0.053$). These cross-sectional results indicate that better cognitive performance was associated most consistently with lower depressive and anxiety symptom burden.

Baseline-adjusted longitudinal associations

Finally, we examined whether baseline cognitive performance showed selective short-term longitudinal associations with later clinical outcomes. Longitudinal analyses were conducted in the paired analytic cohort of 1,056 participants with at least two assessments. This paired analytic cohort was defined using the same core eligibility framework as

the main sample, with outcome-specific complete-case samples used for individual models. For each outcome, follow-up severity was regressed on the corresponding baseline outcome, baseline RBANS Total Scale score, age, sex, education, diagnosis, and follow-up interval, with heteroskedasticity-consistent HC3 standard errors and false-discovery-rate correction applied across the five models (Figure 3B; Table 3). Baseline cognitive performance showed the most robust prospective association with follow-up perceived stress severity. Higher T1 RBANS Total Scale scores predicted lower T2 PSS total after adjustment for baseline PSS and covariates ($\beta = -0.076$, 95% CI $[-0.116, -0.037]$, FDR-adjusted $p < 0.001$). A weaker association was also observed for clinician-rated depressive severity, with higher baseline RBANS total predicting lower T2 HAMD-17 total ($\beta = -0.031$, 95% CI $[-0.057, -0.005]$), although this effect did not remain significant after false-discovery-rate correction (FDR-adjusted $p = 0.052$). No significant prospective associations were observed for T2 PHQ-9 total ($\beta = 0.007$, 95% CI $[-0.017, 0.030]$, FDR-adjusted $p = 0.577$), T2 GAD-7 total ($\beta = -0.007$, 95% CI $[-0.026, 0.012]$, FDR-adjusted $p = 0.577$), or T2 ISI total ($\beta = -0.012$, 95% CI $[-0.033, 0.009]$, FDR-adjusted $p = 0.416$). Taken together, these findings suggest selective rather than broad longitudinal associations of baseline cognitive performance, with the clearest prospective signal observed for subsequent perceived stress severity.

Discussion

This study supports a two-level account of multi-domain psychopathology in adolescents and young adults with affective disorders. At the system level, stress appraisal and sleep disturbance occupied relatively prominent cross-domain structural positions within the joint emotion-cognition association network. At the person level, better cognitive performance was associated with lower current depressive and anxiety symptom burden and, longitudinally, with lower subsequent perceived stress severity. Taken together, these findings suggest that stress and sleep index system-level structural prominence, whereas cognitive performance indexes person-level variation that is expressed cross-sectionally in

affective burden and selectively, over short follow-up intervals, in later perceived stress [10, 13, 14].

Stress appraisal and sleep disturbance as prominent cross-domain structural positions

Across minimum dominating set configurations, control-frequency profiles, and module-level coverage summaries, stress appraisal and sleep disturbance emerged as the most prominent cross-domain components of the estimated association network. Negative stress, in particular, showed the clearest outward structural reach, whereas sleep occupied a secondary but still notable position linking to affective domains. This pattern is broadly consistent with evidence that stress-related processes are strongly implicated in affective symptom expression and that sleep disturbance is both a transdiagnostic correlate and a clinically meaningful predictor of adverse emotional outcomes [7–9, 25–30]. The contribution of the present study is therefore not to re-establish the importance of stress or sleep per se, but to quantify their placement within a joint emotion-cognition association network using structural-positioning metrics that describe repeated minimal coverage and module-level asymmetry.

At the same time, these findings should be interpreted within clear conceptual bounds. The estimated network captures conditional association structure, and domination-based metrics quantify structural coverage within that structure. Accordingly, the prominence of stress and sleep should not be interpreted as evidence of causal control, dynamical controllability, or intervention efficacy. Rather, these findings identify domains that are especially well-positioned within the current estimated system and thereby generate more focused clinical hypotheses about multi-domain organisation while remaining agnostic about the causal mechanisms that produce such positioning [18, 19, 35].

Continuous cognitive performance and current symptom burden

A major revision of the present study was to treat cognitive performance as a continuous dimension rather than to rely on sample-derived dichotomisation. Within this framework, better cognitive performance was associated most consistently with lower depressive and anxiety symptom burden, whereas associations with perceived stress and insomnia were weaker and did not survive false-discovery-rate correction. This pattern is consistent with the broader literature showing that cognitive difficulties in mood disorders are closely linked to current affective burden, including both depressive and anxious symptom dimensions, but are not necessarily reducible to a single categorical subgroup distinction [4–6, 24]. Conceptually, these findings support the value of modelling cognition as a graded clinical characteristic embedded within a broader symptom system, rather than assuming that its role is best captured by a binary impaired-versus-non-impaired contrast.

This shift in analytic emphasis also changes how cognition is positioned within the broader emotion-cognition system. Instead of asking whether a putatively impaired subgroup shows a distinct network organisation, the revised analyses ask whether variation in cognitive performance is systematically related to symptom burden across the full clinical sample. The answer appears to be yes, particularly for depression and anxiety. In this sense, cognition functions less as a grouping variable defining discrete architectures and more as a clinically meaningful person-level dimension associated with current affective severity.

Selective longitudinal associations of baseline cognitive performance

The longitudinal findings were more selective than the cross-sectional results. After adjustment for baseline severity, age, sex, education, diagnosis, and follow-up interval, baseline RBANS Total Scale score showed its clearest prospective association with follow-up perceived stress severity. Higher baseline cognitive performance predicted lower subsequent PSS scores, whereas associations with follow-up PHQ-9, GAD-7, and ISI scores

were not significant after correction, and the association with clinician-rated depressive severity did not survive false-discovery-rate adjustment. This pattern suggests that the most robust short-term prospective correlate of cognitive performance in the present dataset is perceived stress rather than broad downstream improvement across all symptom domains.

Importantly, this longitudinal result should not be framed as a validation of domination-based metrics or as confirmation of a causal control architecture. The longitudinal models did not test whether network-derived structural-positioning indices themselves predicted future outcomes, nor did they manipulate stress or sleep as treatment targets. Nevertheless, the finding that baseline cognition was most strongly related to later perceived stress is directionally consistent with the prominent structural positioning of stress-related indicators in the cross-sectional network. One plausible interpretation is that perceived stress may be especially sensitive to individual differences in cognitive efficiency, appraisal, or coping-related processes over short follow-up intervals [24, 25, 31]. Under this interpretation, the longitudinal results add a complementary person-level perspective to the network findings without converting structural prominence into causal proof.

Methodological contribution and interpretive boundaries

Methodologically, the study illustrates how dominance-based metrics can complement more familiar symptom-network descriptors. Rather than replacing centrality-based summaries, measures such as minimum dominating sets, control frequency, and module-level coverage quantify which indicators repeatedly appear in minimal multi-domain coverage configurations and which domains show relatively broader structural reach. In a setting where emotional symptoms, stress, sleep disturbance, and cognition co-occur, this offers an additional descriptive lens for understanding how different parts of the system are positioned relative to one another [11, 17, 32, 33, 35]. At the same time, the interpretive boundary remains essential: such metrics describe structure within an estimated association network and do not, on their own, establish causal leverage or intervention

priority.

Limitations and future directions

Several limitations should guide interpretation. First, the full-sample network was estimated from cross-sectional data and therefore cannot establish temporal ordering or causal directionality [18, 19]. Second, the use of absolute partial correlations removes sign information, limiting inference about whether associations are facilitative, antagonistic, or compensatory. Third, although the revised longitudinal models are substantially stronger than unadjusted change-score comparisons, the follow-up interval was short (median, 22 days), and treatment exposure was not modelled directly. Fourth, the sample was drawn from a single clinical centre and combined participants with MDD and BD, which may obscure diagnosis-specific architectures. Moreover, the longitudinal analyses were conducted in a restricted analytic cohort derived from a broader repeated-assessment database, and model-wise complete-case estimation further reduced sample size for specific outcomes. Finally, RBANS was analysed here as a clinically useful continuous indicator of overall cognitive performance, but questions regarding developmental appropriateness and optimal scaling in younger participants remain important for future work.

Future studies should extend this framework in three directions. First, longer-term longitudinal designs are needed to test whether baseline cognition, stress, and sleep prospectively relate to one another over clinically meaningful time windows. Second, studies incorporating repeated measurements at the symptom level could examine whether stress-related and sleep-related components also show temporal prominence within dynamic network models. Third, intervention studies will be required to determine whether targeting stress appraisal or sleep disturbance produces broader downstream improvements, and whether such effects depend on baseline cognitive performance. More generally, the present findings support a cautious but potentially useful view of multi-domain psychopathology: structural prominence within an estimated association network can generate tractable clinical hypotheses, but those hypotheses still require prospective and

experimental testing [13, 20].

Conclusion

In adolescents and young adults with affective disorders, stress appraisal and sleep disturbance occupied prominent cross-domain structural positions within a joint emotion-cognition association network. Better cognitive performance was associated with lower concurrent depressive and anxiety symptom burden and showed its clearest prospective association with subsequent perceived stress severity. Together, these findings support an exploratory structural-positioning framework in which stress and sleep represent salient system-level components of a broader symptom-cognition architecture, whereas cognitive performance indexes person-level variation linked to both current affective burden and later stress-related outcomes. These results should be interpreted as hypothesis-generating rather than as evidence of causal control or intervention efficacy [18, 19].

Acknowledgments

Ethics

Ethical approval for this study was granted by the Ethics Committee of the Affiliated Nanjing Brain Hospital, Nanjing Medical University (Nanjing, China). Written informed consent was obtained from all participants and their parents or legal guardians following a full explanation of the study procedures.

Conflict of Interest

The authors declare no conflict of interest to this work.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Figure legends

Figure 1. Full-sample network architecture and node-level structural positioning (A) Full-sample joint symptom network estimated using Graphical Lasso. Nodes represent individual indicators, and node colours denote Louvain communities (depression, anxiety, negative stress, positive stress, sleep, and cognition). Edges indicate non-zero absolute partial correlations. (B) Summary of global network properties used to characterise the overall architecture of the estimated network, including the regularisation parameter, number of nodes and edges, density, modularity, and domination-based summary statistics. Global strength in panel B is computed as the node-normalised total absolute edge weight ($\sum_{i < j} |w_{ij}|/N$). (C) Standardised node-property profiles (Z-scores; e.g., strength, clustering, closeness, betweenness, degree, and eigenvector centrality) illustrating heterogeneity in how indicators are positioned within the network topology. (D) Control frequency (CF) Z-scores, where CF quantifies how often an indicator appears in enumerated minimum dominating sets (MDSs), that is, minimal structural-coverage configurations of the network. Here, “control” is retained as part of the original metric name and is not interpreted as evidence of causal control. Dashed lines separate node groups/domains.

Figure 2. Macro-level cross-module structural coverage architecture (MDS/CF/AMCS)

(A) Symptom network with node colour indicating control frequency (CF; see colour bar). Edge width reflects the magnitude of absolute partial correlations. Red numeric labels indicate CF values for selected high-CF indicators. (B) Module in-strength, summarising the total incoming cross-module structural coverage for each module (aggregated across the module-level coverage network). (C) Module out-strength, summarising the total outgoing cross-module structural coverage for each module. (D) Average control frequency aggregated at the module level. Bar colours denote module groups (Peripheral Emotional Modules, Intrinsic Emotional Modules, Cognitive Modules). (E) Directed module-level backbone derived from Average Module Control Strength (AMCS), highlighting dominant cross-module structural coverage pathways; edge labels indicate corresponding

AMCS weights. (F) Heatmap of AMCS between module groups. Cell values and colour intensity represent the magnitude of cross-group structural coverage, summarising the hierarchical coupling between affective and cognitive domains.

Figure 3. Adjusted associations of cognitive performance with cross-sectional symptom burden and longitudinal outcomes (A) Cross-sectional adjusted associations between RBANS Total Scale score and symptom burden. Points indicate regression coefficients and horizontal bars indicate 95% confidence intervals from linear regression models predicting PHQ-9, GAD-7, PSS, and ISI total scores from RBANS Total Scale score, adjusted for age, sex, education, and diagnosis. (B) Baseline-adjusted longitudinal associations between T1 RBANS Total Scale score and T2 outcomes. Points indicate regression coefficients and horizontal bars indicate 95% confidence intervals from linear regression models predicting T2 HAMD-17, PHQ-9, GAD-7, PSS, and ISI total scores from T1 RBANS Total Scale score, adjusted for the corresponding T1 outcome, age, sex, education, diagnosis, and follow-up interval. For both panels, negative coefficients indicate that higher cognitive performance was associated with lower symptom burden. Right-side columns report the estimated coefficient with its 95% confidence interval, model sample size, adjusted R^2 , and false-discovery-rate (FDR) adjusted p value. Asterisks indicate FDR-significant associations. Longitudinal estimates are based on outcome-specific complete-case samples from the paired analytic cohort.

Table legends

Table 1: Sample characteristics of the cross-sectional and longitudinal analytic cohorts

Panel A. Cross-sectional sample characteristics	
Analytic sample size, n	1,027
Age, mean (SD), years	15.34 (2.03)
Female, n (%)	739 (72.0)
Male, n (%)	288 (28.0)
Major Depressive Disorder, n (%)	531 (51.7)
Bipolar Disorder, n (%)	496 (48.3)
Cross-sectional continuous regression sample, n ^a	1,024
Panel B. Longitudinal analytic cohort characteristics	
Paired analytic cohort size, n	1,056
Baseline age, mean (SD), years	15.38 (2.05)
Female, n (%)	760 (72.0)
Male, n (%)	296 (28.0)
Baseline Major Depressive Disorder, n (%)	561 (53.1)
Baseline Bipolar Disorder, n (%)	495 (46.9)
Follow-up interval, median (IQR), days	22 (15–35)
Longitudinal HAMD-17 model sample, n ^a	1,031
Longitudinal PHQ-9 model sample, n ^a	982
Longitudinal GAD-7 model sample, n ^a	982
Longitudinal PSS model sample, n ^a	965
Longitudinal ISI model sample, n ^a	981

For cross-sectional analyses, participants were included only if complete data were available for the symptom outcome, RBANS Total Scale score, age, sex, education, and diagnosis. For longitudinal analyses, participants were included only if complete data were available for the corresponding T1 and T2 outcome measures, T1 RBANS Total Scale score, age, sex, education, diagnosis, and follow-up interval. The longitudinal analytic cohort shown in Panel B was derived from a broader repeated-assessment source cohort in the raw database.

Table 2: Cross-sectional adjusted associations of RBANS total with symptom burden

Outcome	N	β for RBANS total	Robust SE (HC3)	95% CI	p	FDR-adjusted p	Adjusted R^2
PHQ-9 total	1,024	-0.029	0.010	-0.048 to -0.011	0.002	0.004	0.037
GAD-7 total	1,024	-0.037	0.008	-0.053 to -0.021	< 0.001	< 0.001	0.039
PSS total	1,024	-0.026	0.013	-0.052 to 0.0003	0.053	0.053	0.028
ISI total	1,024	-0.018	0.009	-0.036 to -0.001	0.041	0.053	0.024

Linear regression models were adjusted for age, sex, education, and diagnosis. Robust standard errors were estimated using the HC3 estimator. False discovery rate (FDR) correction was performed across the four symptom outcomes using the Benjamini–Hochberg procedure. Negative coefficients indicate that higher RBANS Total Scale scores were associated with lower symptom burden.

Table 3: Baseline-adjusted longitudinal associations of T1 RBANS total with T2 outcomes

Outcome	N	β for T1 RBANS total	Robust SE (HC3)	95% CI	p	FDR-adjusted p	Adjusted R^2
HAMD-17 total	1,031	-0.031	0.013	-0.057 to -0.005	0.021	0.052	0.067
PHQ-9 total	982	0.007	0.012	-0.017 to 0.030	0.577	0.577	0.163
GAD-7 total	982	-0.007	0.010	-0.026 to 0.012	0.480	0.577	0.140
PSS total	965	-0.076	0.020	-0.116 to -0.037	< 0.001	< 0.001	0.133
ISI total	981	-0.012	0.011	-0.033 to 0.009	0.249	0.416	0.137

Linear regression models predicted each T2 outcome from T1 RBANS Total Scale score while adjusting for the corresponding T1 outcome, age, sex, education, diagnosis, and follow-up interval. Robust standard errors were estimated using the HC3 estimator. False discovery rate (FDR) correction was performed across the five longitudinal outcome models using the Benjamini–Hochberg procedure. Negative coefficients indicate that better baseline cognitive performance was associated with lower symptom burden at follow-up after covariate adjustment.

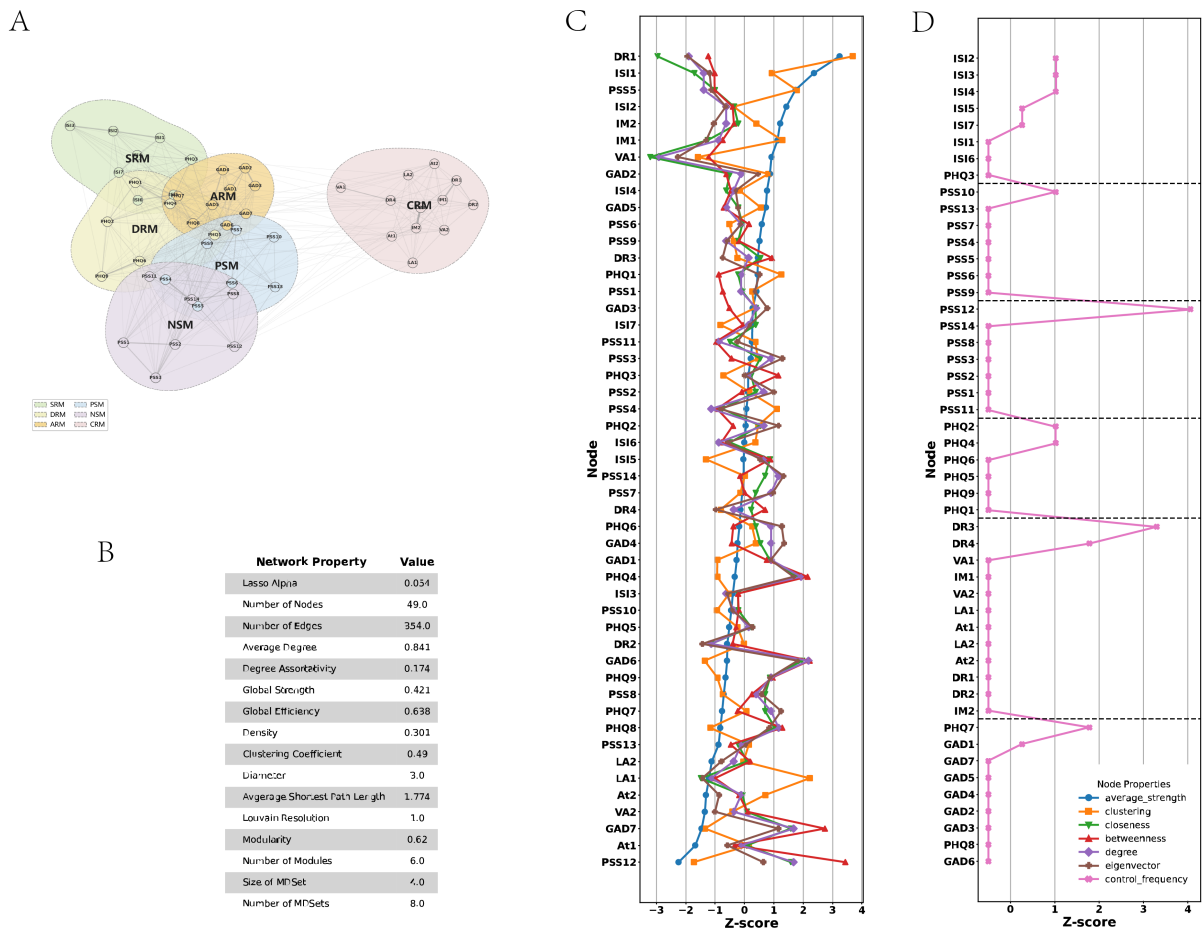


Figure 1

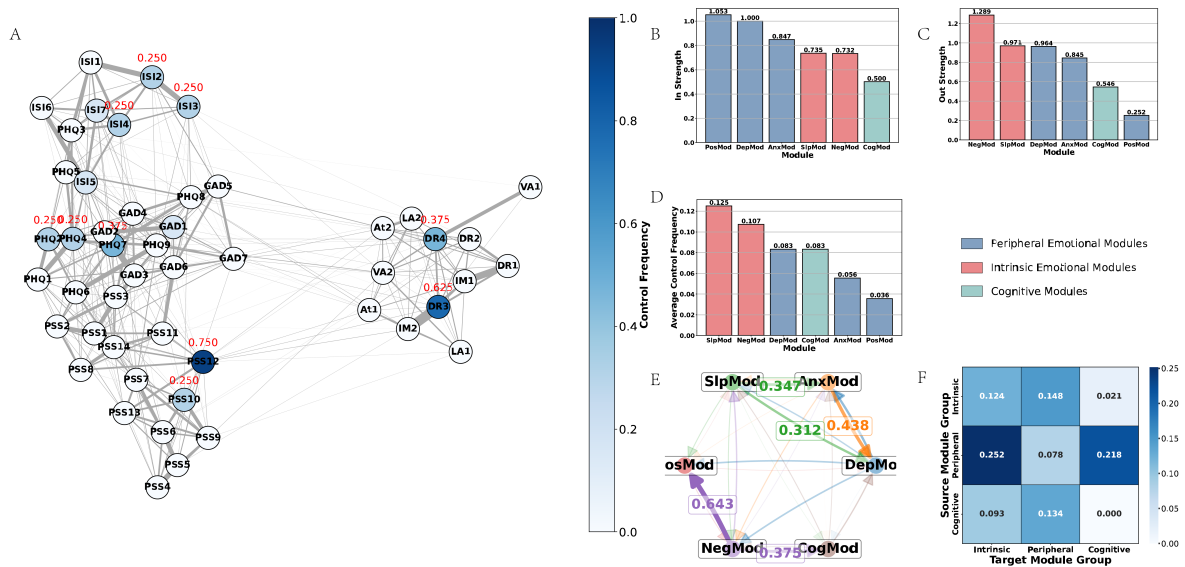


Figure 2

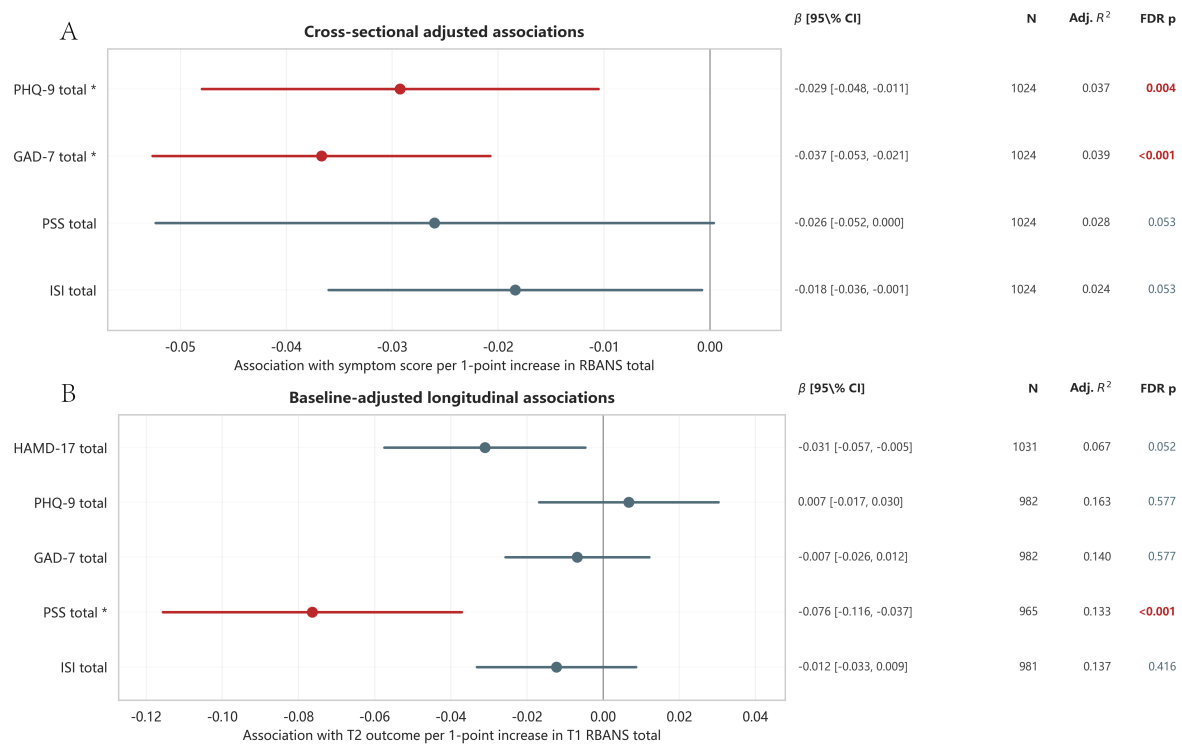


Figure 3